AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I130*

Team Name: *TECH ENTITY*

Domain: *6-Healthcare, Biotechnology and Human Well-being*

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Problem Statement

THE PROBLEM WE AIM TO SOLVE: In critical-care units, mechanically ventilated patients can experience rapid changes in respiratory condition, requiring continuous monitoring and timely ventilator adjustments. Clinicians managing multiple patients may not always be able to respond immediately to every developing change, potentially delaying intervention. There is a need for an intelligent system that predicts early signs of respiratory deterioration using existing patient and ventilator data and enables timely, safety-constrained automatic adjustments while keeping clinicians in supervisory control.

Problem Understanding

- **Who experiences this problem?**
  - Mechanically ventilated patients in critical-care units.
  - Doctors and nurses responsible for continuous patient monitoring.

- **What causes the problem?**
  - Rapid changes in patients' respiratory conditions.
  - Difficulty in continuously monitoring multiple patients.
  - Delayed recognition of early deterioration patterns.

- **How does it affect users or stakeholders?**
  - Delayed intervention when a patient's condition worsens.
  - Increased workload for healthcare professionals.
  - Potential risk of serious respiratory complications.

Existing Gaps and Challenges

- **Existing Gap or Limitation 1**
  - Early deterioration can be difficult to identify from individual readings.
  - Many automated systems respond to current patient conditions.
- **Existing Gap or Limitation 2**
  - Advanced ventilation automation may be limited by cost and compatibility.
  - Continuous monitoring increases clinicians' workload.
- **Main Challenge That Remains Unsolved**
  - **Predicting deterioration early enough for timely, safe ventilator adjustments while maintaining clinician supervision.**

Proposed Solution

Core Idea or Product

- An AI-based predictive ventilation system compatible with existing ventilators.
- Predicts respiratory deterioration using existing patient and ventilator data.

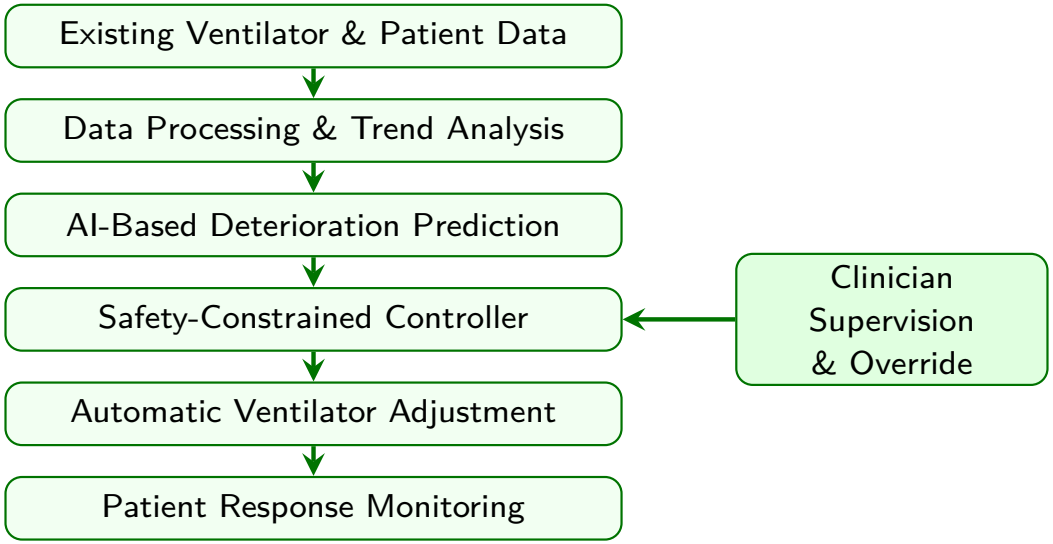
How the Solution Works

- Continuously analyzes patient data to identify early deterioration patterns.
- Predicts deterioration and enables safety-constrained automatic ventilator adjustments.
- Monitors patient response while keeping clinicians in supervisory control.

Main Value Delivered to Users

- Enables earlier intervention to help prevent worsening respiratory conditions.
- Reduces the burden of continuous manual monitoring.
- Supports responsive, personalized respiratory care.

Technical Approach



AI predicts respiratory deterioration from existing data. A safety controller checks permitted adjustments, while patient responses are monitored and clinicians retain override control.

Tech Stack

Frontend: React.js

Backend: Python, FastAPI

Database: PostgreSQL

AI/ML, APIs, or Hardware: Scikit-learn, XGBoost, Ventilator Interface (future)

What Is New or Different?

Predictive AI

- Predicts potential respiratory deterioration by analyzing patient data trends.
- Aims to act before significant deterioration occurs rather than simply reacting to changes.
- Analyzes patterns across multiple patient parameters to identify early warning signs.

Why Is This Approach Suited to the Problem?

- Continuously analyzes available patient and ventilator data.
- Supports timely adjustments while clinicians manage multiple patients.
- Considers individual patient trends to support personalized respiratory care.

What Is the Key Differentiator?

- Combines predictive AI, automatic ventilator adjustment, safety constraints, and clinician supervision.
- Explores integration with compatible existing ventilators, potentially reducing the need for costly equipment replacement.

Feasibility, Scalability and Risks

Feasibility

- Use public ICU datasets such as MIMIC-IV to train and evaluate the AI model.
- Develop a prototype using Python, machine learning, and a user-friendly dashboard.
- Demonstrate predictive ventilation adjustments using simulated patient data.

Scalability

- Expand the system to monitor multiple patients across critical-care units.
- Explore integration with different compatible ventilator models.
- Improve predictions using additional validated clinical data.

Risks and Mitigation

- **Limited Data:** Use reliable datasets and validate data quality.
- **Incorrect Predictions:** Evaluate model accuracy and minimize false alarms.
- **Patient Safety:** Enforce safety limits and provide clinician override.
- **Device Compatibility:** Start with simulation and use manufacturer-supported interfaces.

Estimated Project Cost

- **Development or Hardware Cost: 5,000–10,000**
  - AI Model Development: 0 using Python and open-source libraries.
  - Dashboard Development: 0 using free development tools.
  - Testing and Simulation: 1,000–3,000.
  - Computing and Miscellaneous Expenses: 4,000–7,000.
  - No new ventilator hardware is required for the initial prototype.

- **Hosting or Maintenance Cost: 0–2,000/month**
  - Cloud Hosting: 0–1,000/month.
  - Database and Storage: 0–500/month.
  - Maintenance and Updates: 0–500/month.

- **Estimated Total**
  - Initial Development: 5,000–10,000.
  - First-Year Estimated Cost: 5,000–34,000.

Note: Estimates are for an ideathon prototype using public ICU datasets, open-source tools, and simulated ventilator data. Real-world integration and clinical validation would require additional funding.

Use Cases Overview

1 Early Deterioration Prediction

- Analyzes existing patient data
- Identifies early respiratory changes
- Generates risk alerts

2 Safety-Constrained Adjustment

- Evaluates predicted risk
- Checks predefined safety limits
- Supports controlled adjustments

3 Response Monitoring

- Monitors patient response
- Reassesses risk using new data
- Alerts clinicians when needed

Project Overview

An AI-based predictive ventilation concept that analyzes existing patient and ventilator data, predicts respiratory deterioration, and supports safety-constrained adjustments under clinician supervision.

System Workflow

Existing Data → AI Prediction → Safety Controller → Ventilator Adjustment → Response Monitoring

Impacts and Benefits

Improved Patient Care

- Identifies early signs of respiratory deterioration to support timely intervention.

Predictive and Adaptive Ventilation

- Analyzes existing patient and ventilator data to predict respiratory changes and enable safety-constrained automatic adjustments.

Reduced Monitoring Burden

- Assists ICU professionals while maintaining clinician supervision and override control.

Measurable System Performance

- Evaluates prediction accuracy, early-warning time, false-alarm rate, and the safety of simulated adjustments.

Compatibility With Existing Equipment

- Aims to use existing monitoring infrastructure without additional sensors, subject to compatibility and validation.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 3	3.8	Good Health and Well-Being	Supports timely intervention and improved critical-care support.
SDG 9	9.5	Industry, Innovation and Infrastructure	Promotes AI-driven innovation in existing medical equipment.

Individual Member Contribution

Member	Contribution or Responsibility
Mithun Karthik Dhaneshkumar	Research paper review and problem identification
Balakrishnan S	Research on datasets and potential data sources
Manju Bhashini Senthilkumar	Exploration of AI prediction and system development approaches
Thiyaasri M	System workflow design and solution feasibility analysis

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